

# USING VIRTUAL REALITY TO STUDY FREEZING OF GAIT IN PARKINSON DISEASE



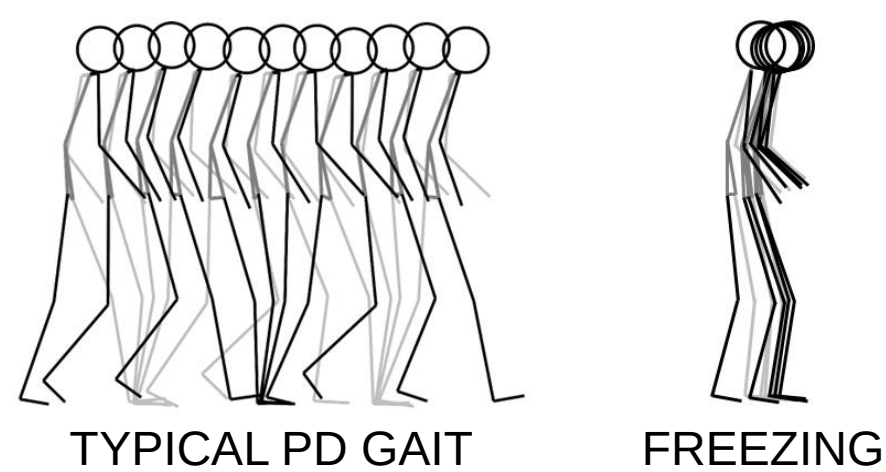
Anjali Singh,<sup>a</sup> Tasnim Alam,<sup>a</sup> Kianna Bolante,<sup>a</sup> Meghna Shankar,<sup>b</sup> Dana Christie,<sup>b</sup> Momona Yamagami,<sup>c</sup> Sujata Pradhan,<sup>b</sup> Valerie E. Kelly<sup>b</sup>

<sup>a</sup>Paul G. Allen School of Computer Science and Engineering and <sup>b</sup>Department of Rehabilitation Medicine, University of Washington, Seattle, WA,  
<sup>c</sup>Department of Electrical and Computer Engineering, Rice University, Houston, TX

## BACKGROUND

**FREEZING OF GAIT** is a transient disturbance of walking, where people cannot take a step despite trying to do so.

- Freezing is **common in people with advanced Parkinson's disease (PD)**<sup>1,2</sup>
- Freezing contributes to **increased fall risk, fall-related injuries, and worse quality of life**
- Though freezing is common in daily life, it is **difficult to elicit and manage** in research and clinical settings



**VIRTUAL REALITY (VR)** has the potential to:

- Simulate real-life environments** that trigger freezing while reducing physical barriers to safety
- Support implementation of **personalized visual cues** that help reduce freezing
- Enable dynamic manipulation** of virtual elements to study causes of freezing

**OBJECTIVE:** Iteratively co-create and evaluate environment CREATION, AUTOMATION, and MANIPULATION tools for a VR platform to study, assess, and treat freezing of gait.

## ENVIRONMENT CREATION

**GOAL:** Simulate real-world environments that provoke freezing

### RATIONALE:

- Narrow spaces, doorways, and dim lighting** can cause freezing<sup>5</sup>
- VR can **replicate** these features without physical barriers to **increase safety**

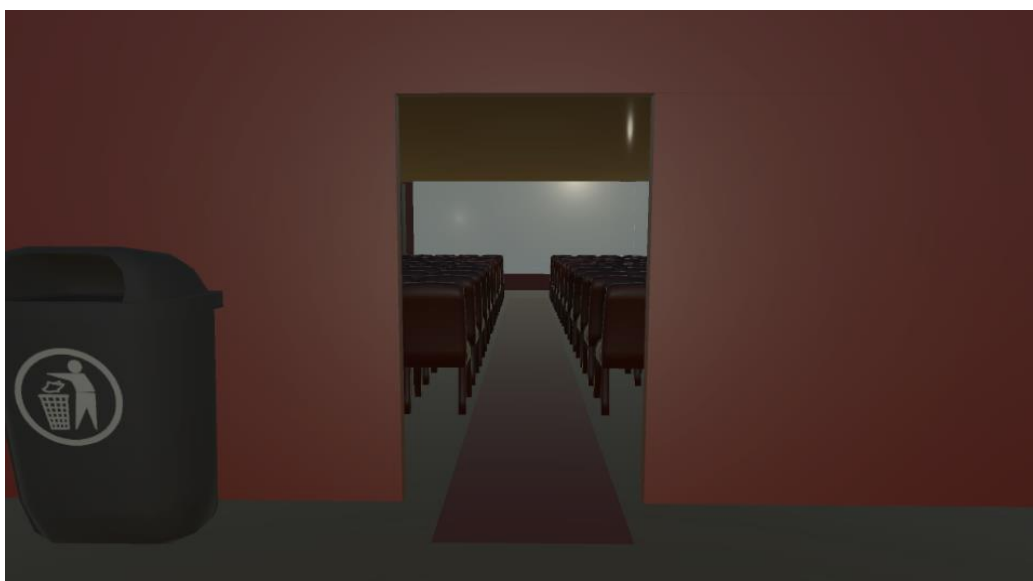
### METHODS:

- Input from people with PD, clinicians, research for **environment selection**
- Environments created in **Unity** and displayed with **HTC Vive headset**
- Participants walked in VR environments & provided feedback to refine

### RESULTS:

- Developed and refined movie theater and hotel hallway environments

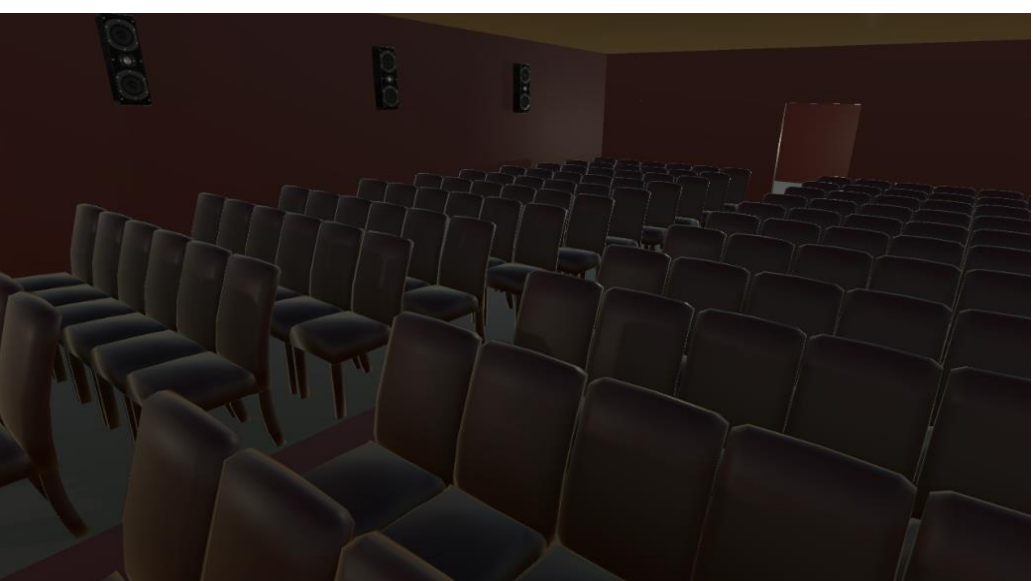
#### MOVIE THEATER



Lobby, looking into movie theater

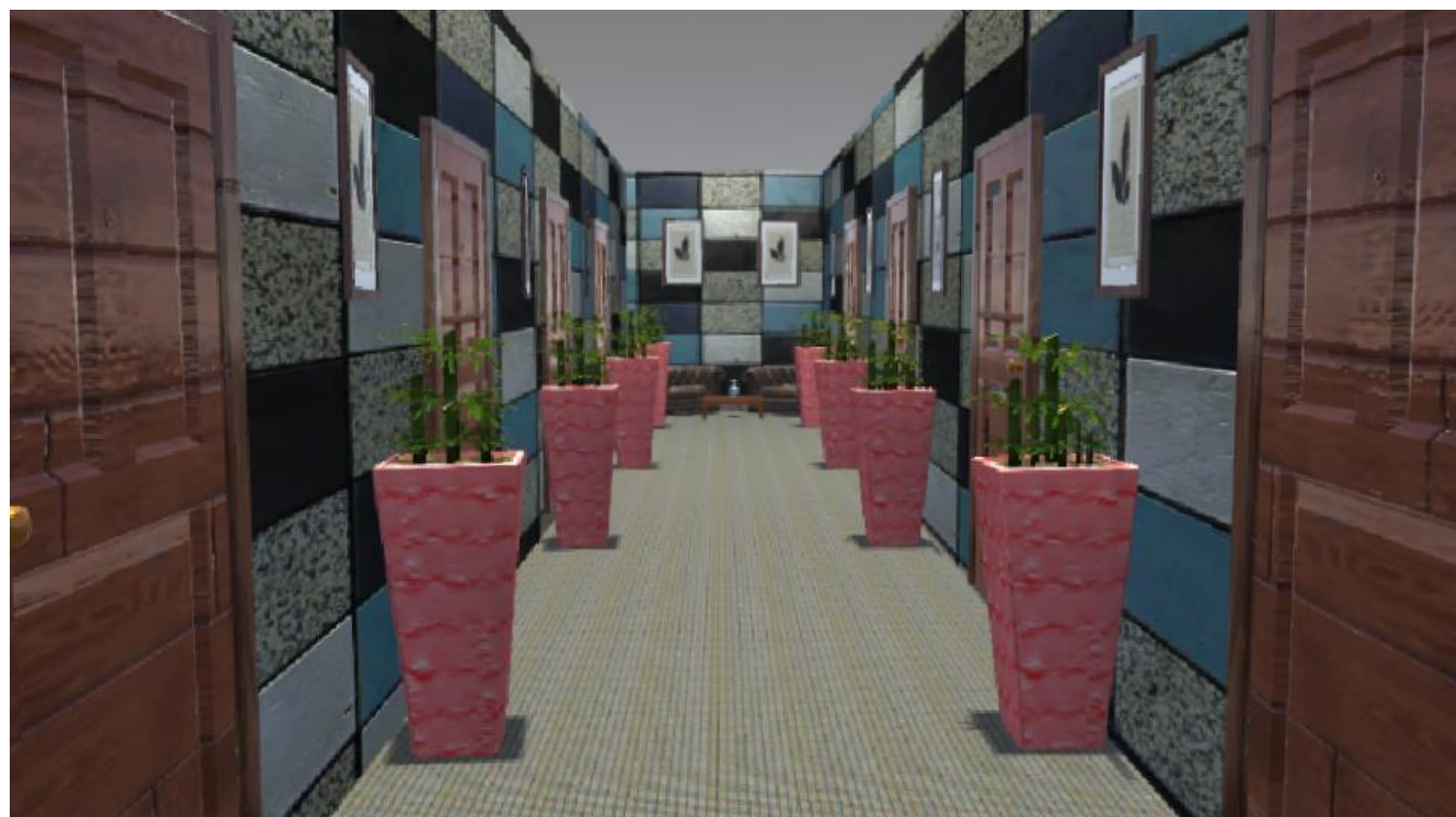


Walking through movie theater



View from front of movie theater

#### HOTEL HALLWAY



View from end of hallway

**Feedback from people with PD and clinicians:**

#### Supported environment design:

- Moving into a darker room beyond a doorway is a common freezing trigger<sup>9</sup>
- "Moving from **open to narrow spaces** triggers **imbalanced gait**"  
– Neurologic Physical Therapist

#### Suggested future enhancements:

- Busy patterns on floors and walls
- Theater walls that angle out are disorienting
- Finding a specific theater seat adds motor and cognitive challenge
- Passages off hotel hallway add complexity

## ENVIRONMENT AUTOMATION

**GOAL:** Provide personalized visual cues for treatment of freezing

### RATIONALE:

- In clinical settings, lines of tape are **placed on the ground manually** to create **visual cues** that are effective to reduce freezing<sup>6,7</sup>
- VR allows **efficient and personalized** generation of cues in **any environment**

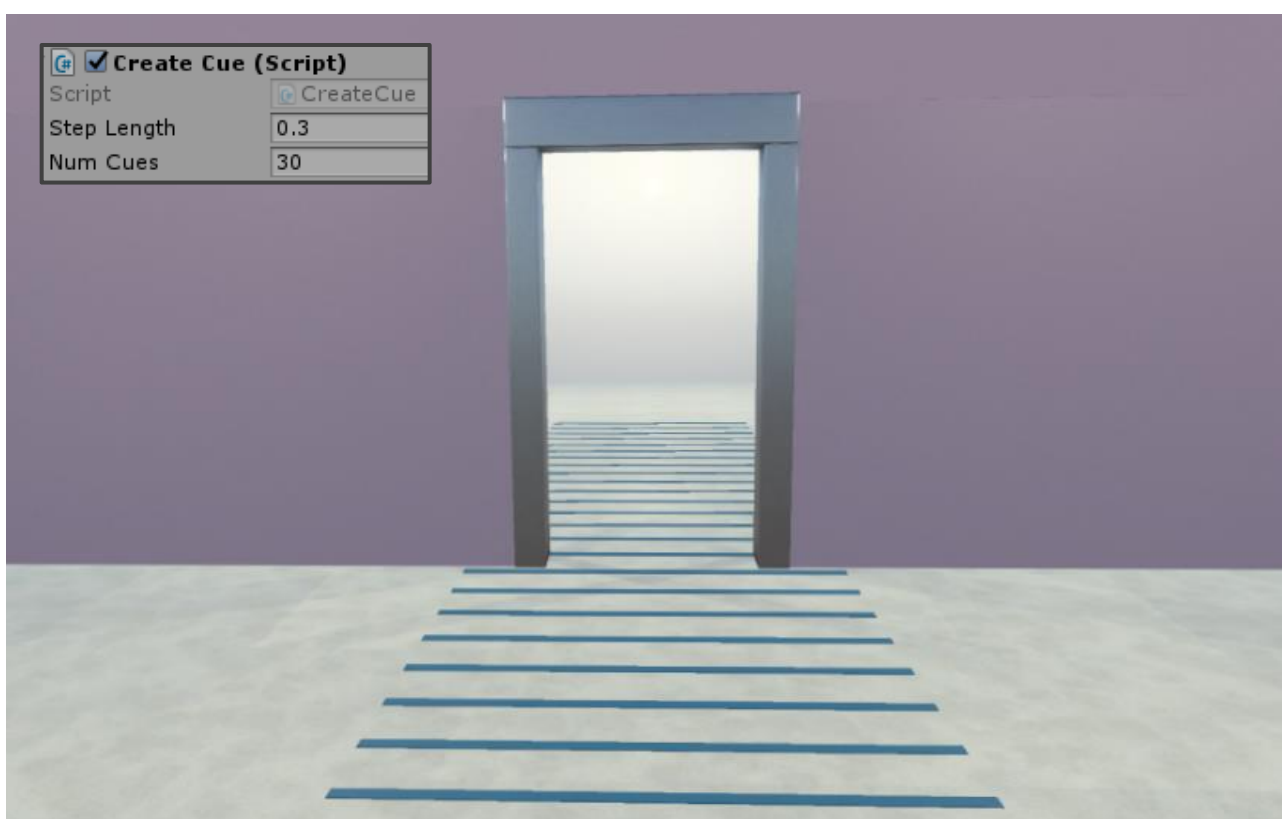
### METHODS:

- Developed **C# scripts** to **dynamically automate** cues (origin, quantity, spacing)
- Using **Qualisys motion capture** to quantify effects on gait pattern

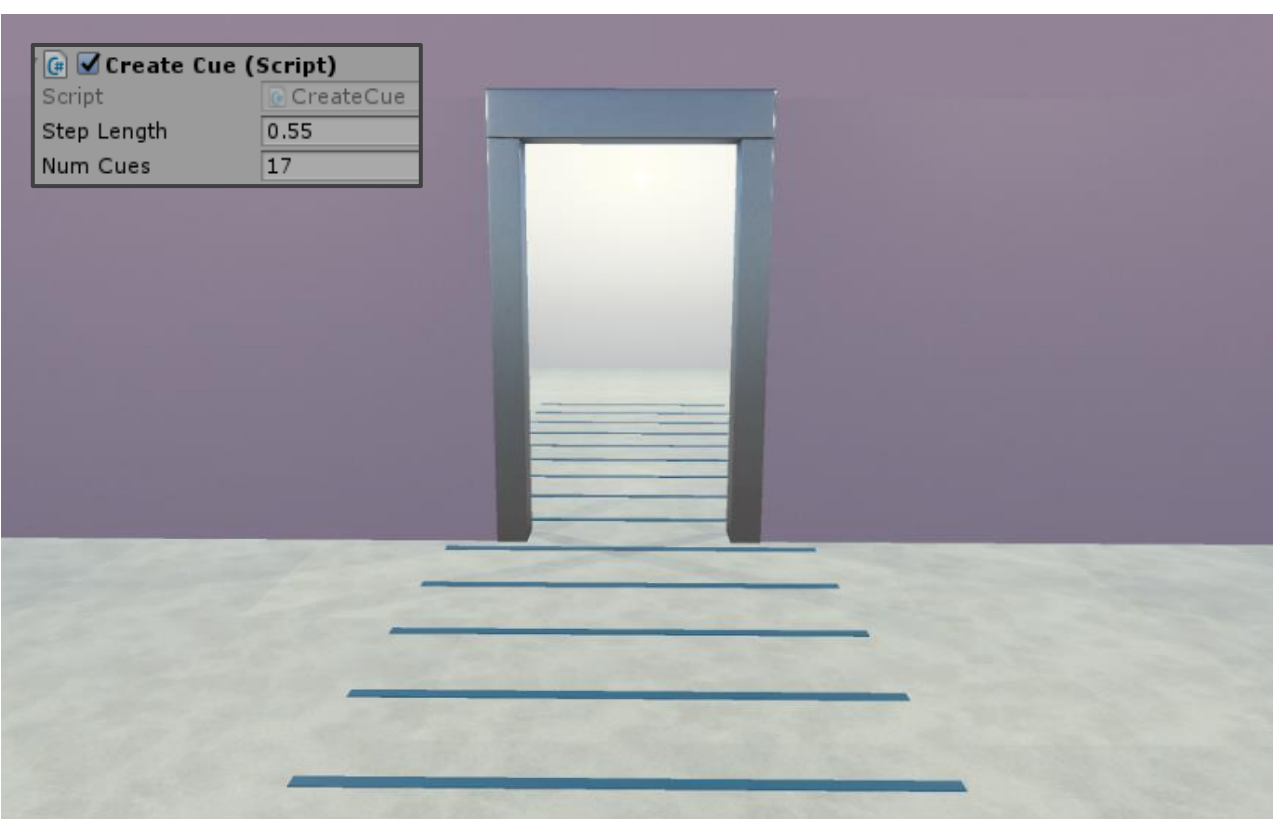
### RESULTS:

- Implementation of cues** in doorway environment (tested with and without cues)

#### VIRTUAL CUE IMPLEMENTATION

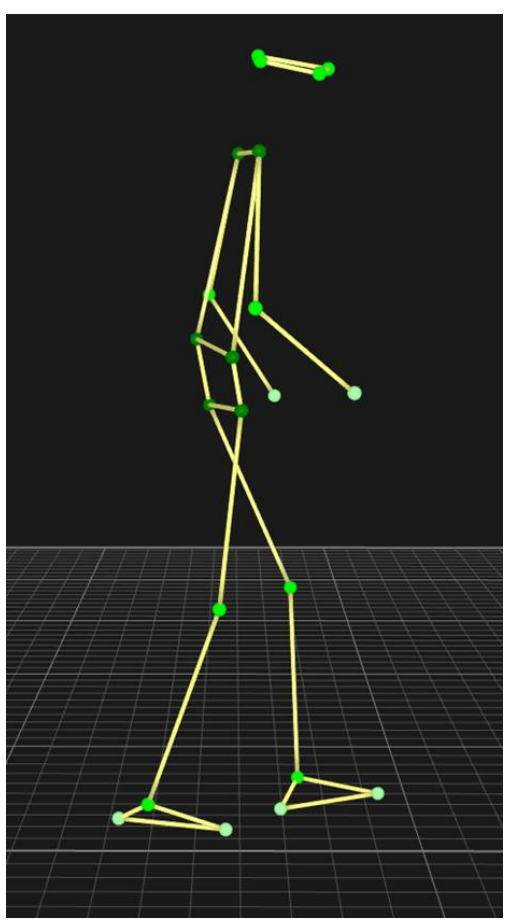


Cue length: 0.3 m

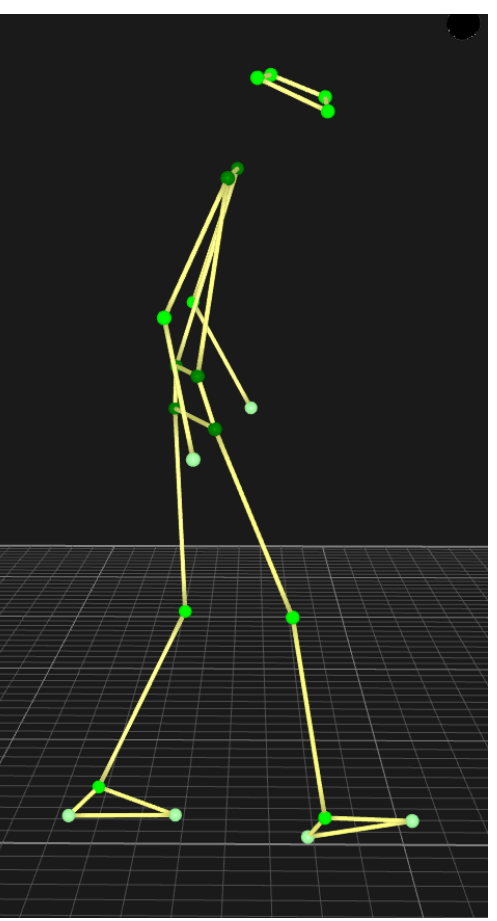


Cue length: 0.55 m

#### MOTION CAPTURE



No cue



With 0.3 m cue

#### Feedback from clinicians:

- Clinicians value** efficiency and ease-of-use for implementing cues
- Ability to **individualize and dynamically apply** in **any environment** is a strength

#### Future enhancements:

- Implement **radial cues for turning** around, which often provokes freezing
- Ongoing testing to determine **effects of cues** for people with PD
- Scripts to **change cues** in response to step-by-step changes

## ENVIRONMENT MANIPULATION

**GOAL:** Create visual perturbations to study mechanisms of freezing

### RATIONALE:

- People with PD have **difficulty using visual inputs** for self-motion perception<sup>8</sup>
- Moving VR environments** can shed light on **sensory contributions to freezing**

### METHODS:

- Created a **virtual visual surround box** with patterned walls
- Developed **C# scripts** to **rotate & translate** surround (variable direction, magnitude, speed)

### RESULTS:

- Pilot testing of visual perturbations in healthy adults and people with PD

#### STATIC SURROUND



#### ROTATE SURROUND

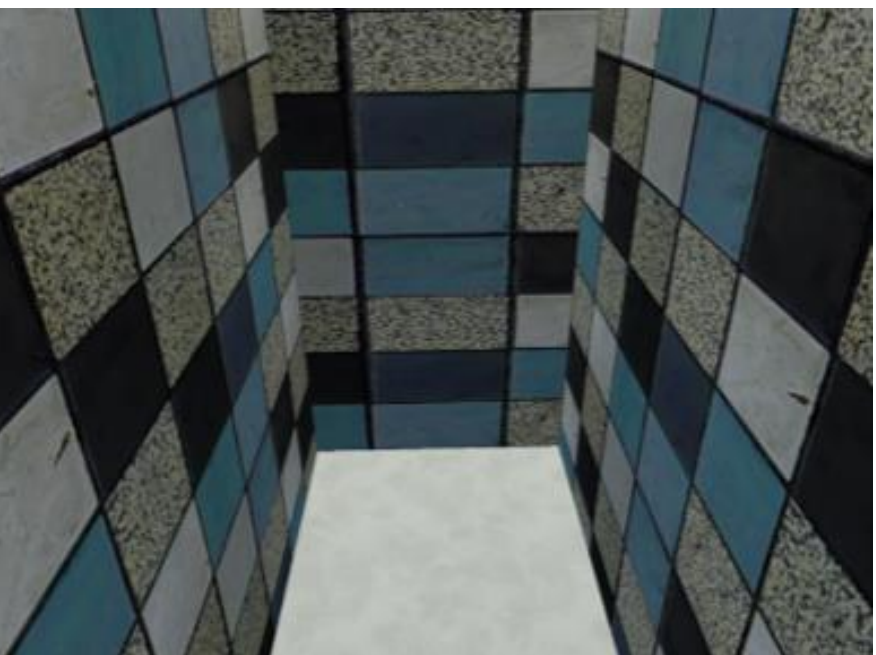
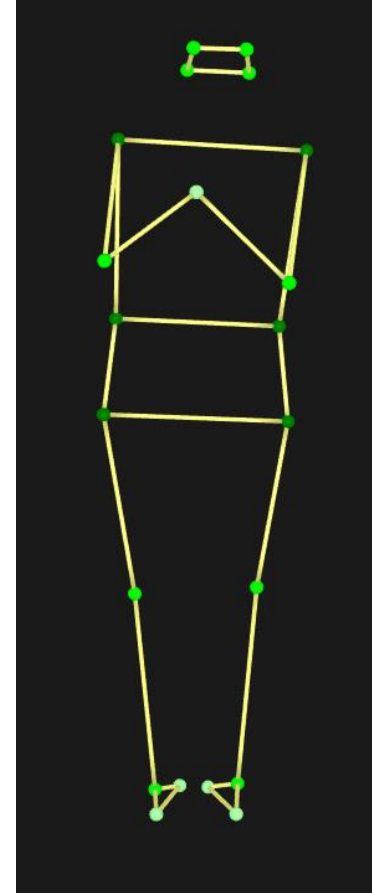


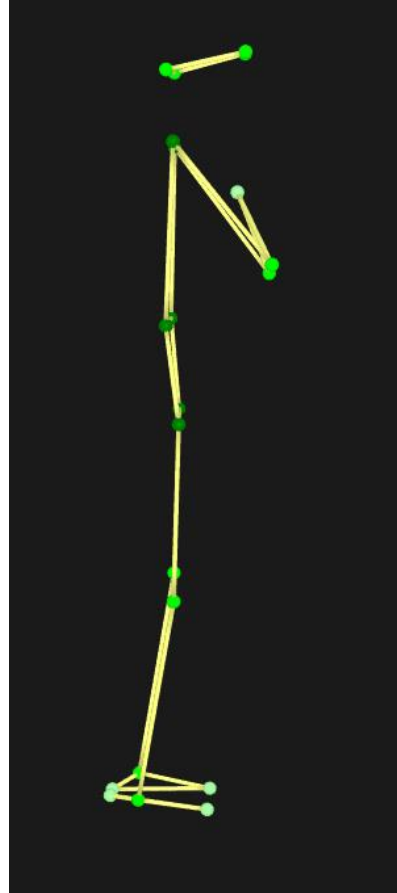
Table. Anterior/posterior center of mass excursion (cm)

|                             | Healthy adults (n=4) | People with PD (n=2) | Example from PD participant: Trace of sagittal plane center of mass excursion |
|-----------------------------|----------------------|----------------------|-------------------------------------------------------------------------------|
| Static surround             | 2.2                  | 2.3                  |                                                                               |
| Rotational perturbations    | 2.4                  | 1.9                  |                                                                               |
| Translational perturbations | 2.6                  | 2.6                  |                                                                               |

#### MOTION CAPTURE



Frontal view



Sagittal view

#### Future enhancements:

- Continue pilot testing of current protocol
- Implement **rotational and mediolateral** perturbations

#### QR CODE



LEARN MORE!

## CONCLUSION & FUTURE ENHANCEMENTS

- Freezing is prevalent among people with PD but remains poorly understood with **limited technology treatment options**.
- We developed a VR platform, with **strong potential** for use in research and clinical settings, that can **advance our understanding of the mechanisms of freezing and optimal treatments**.
- Ongoing work to guide future development includes **qualitative perspectives from people with PD and clinicians** as well as **qualitative assessment of how VR environments, cues, and manipulations affect gait** in people with PD.

## ACKNOWLEDGEMENTS

We wish to thank our participants for their involvement and the Washington State Parkinson Disease Registry for assistance with recruitment.



## REFERENCES

- Nutt, *Lancet Neurol*, 2011
- Weiss, *Brain*, 2020
- Yamagami, *Dis Rehab*, 2022
- Besharat, *Gait Post*, 2023
- Rahman, *Behav Neurol*, 2008
- Osborne, *Phys Ther*, 2022
- Gilat, *npj Park Dis*, 2021
- Yakubovich, *Brain Comm*, 2020
- Cockx, *Front Neurol*, 2023